SageML:

Golf Club Selection Tool

SAGEML, Machine Learning

Stephen Kwak, Federica Domecq

Why golf?

*pst, this was yesterday!

Stephen Kwak

Plays baseball.
Attempting to break 100 in golf.

Fede Domecq

Plays golf well... sometimes.
Attempting to play some better golf.

Statistics in Golf

- Moneyball:
  - Statistics focused approach to baseball recruiting
- Difficulties in golf:

  No shot is ever the same:
  - Distance, lie, weather conditions, pin position

  Club selection:
  - Distance & accuracy
  - Shot lie & hazards


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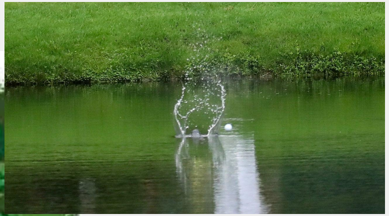
1

Rough

Water hazard



Fairway

Green

Bunker, Sand trap

ML in Golf

Golf Club Selection with AI-Based Game Planning

(Mehdi Khazaeli & Leili Javadpour)

About: Using AI to recommend club based on shot lie, distance to hole, and environmental factors (27 dimensions)

Notable Facts:

- Only took player's 7-iron data into account
- Measured probability of birdie, par, bogey & worse
- Deep learning model performed best (72% accuracy)
- Incorporation of terrain and environmental data decreased entropy error

Existing strategy apps/statistical models

Golfmetrics
Mark Broadie, Columbia

- Strokes gained for each shot relative to PGA/LPGA/NCAA baselines
- Helps players assess strengths and weaknesses by shot category

Performance tracker

Upgame
Sameer Sawhney

- Uses strokes gained as well
- Visual dashboards to help golfers reflect on tendencies

Performance tracking and course strategy app

SG ≈ Expected - Actual

Our Approach

1) Model Expected strokes to hole out (ESHO) from historic PP data within 70 yards of a reachable target
2) Use trackman data (3d tracker) of different club data and “drop” that on top of our target’s ESHO distribution
3) calculate expected strokes for different clubs + aimpoint
4) Penalise if need be with loss function (special circumstance)
5) Give recommendation

Datasets

- Federica’s personal Trackman data for shot dispersion
  - Limited entries → would be what the user presents
- Pomona-Pitzer’s Mens & Womens Tournament data (120k entries) → used to build ESHO

Assumptions

- Recorded types of swings → full swing/half swing
- No wind
- Start from “fair” lie, (i.e. tee, fairway or shot rough acting as fairway)
- No obstacles in the way (i.e. tree, or anything to curve around)
- Carry distances of a relatively consistent golfer

Modelling ESHO using GPR

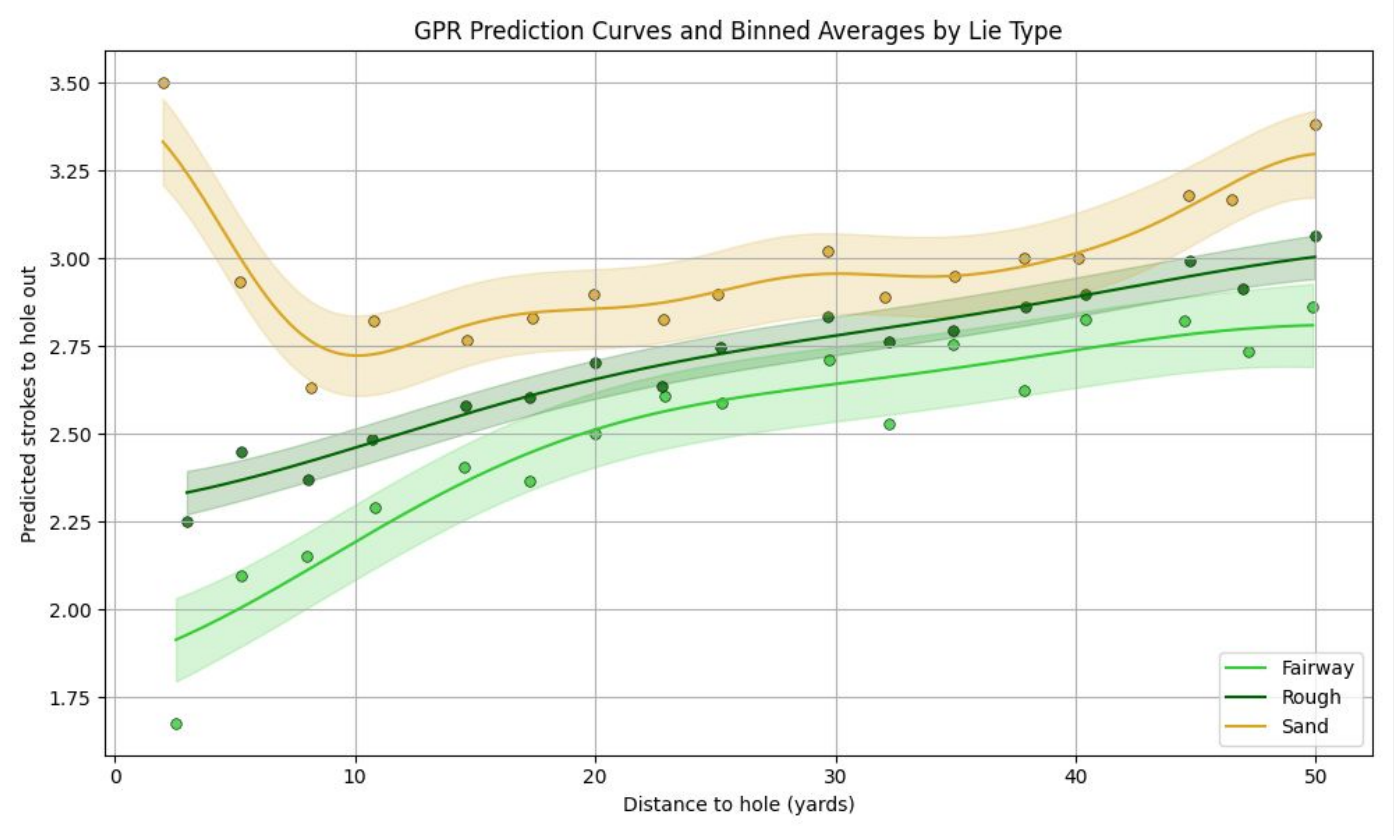
- Gaussian Process Regression
  - 4 different terrains: green, fairway, rough, sand
- What is GPR?

  Non-parametric, flexible regression technique.

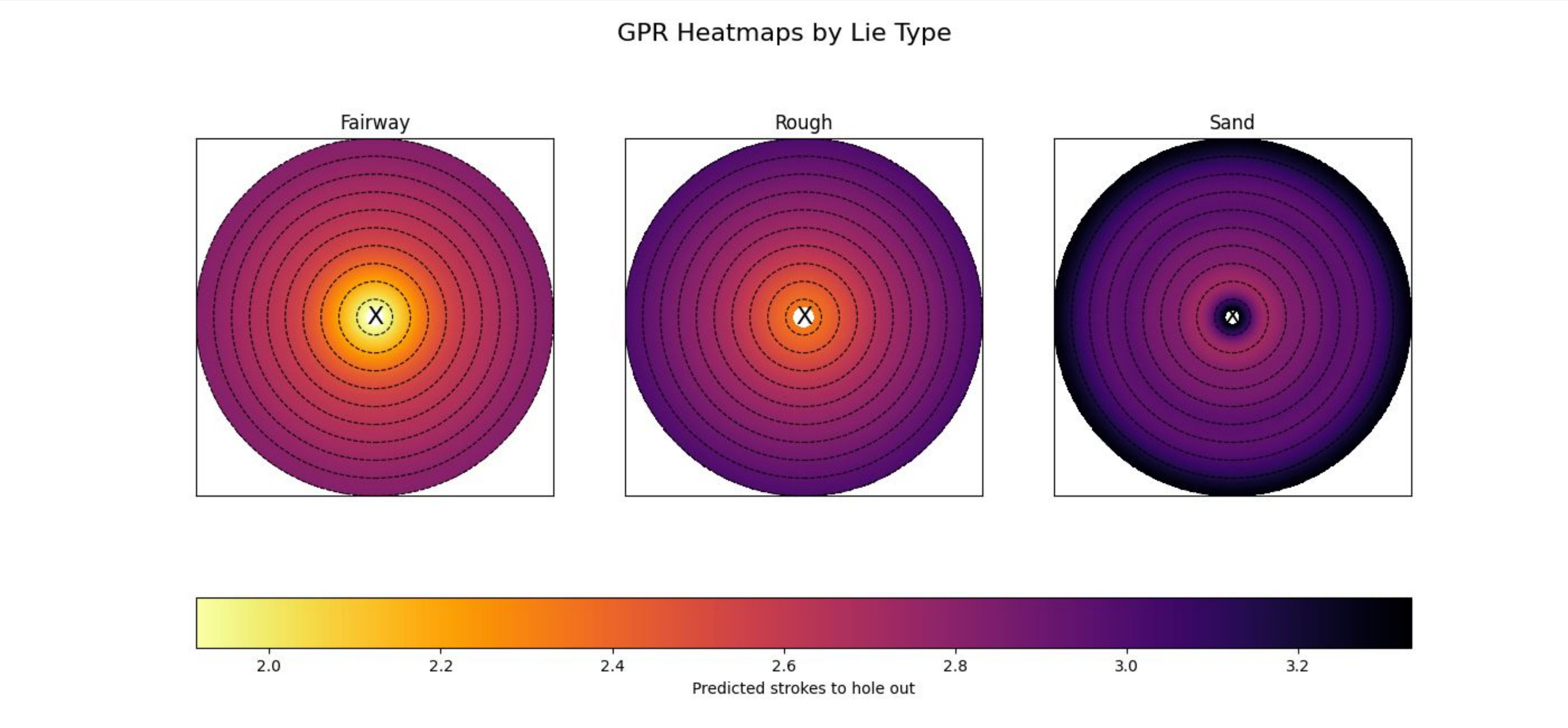
  Models complex relationships without assuming a specific function form.

  Great for capturing smooth patterns.

1) Fit smooth data without assuming specific shape
2) considers all possible smooth curves that could go through or near data, chooses most likely given what's been seen
3) choose? compares each pair of data points with a kernel function (how similar two inputs are) -> final prediction at a new point is a weighted average of all the known outputs where nearby points have more influence.

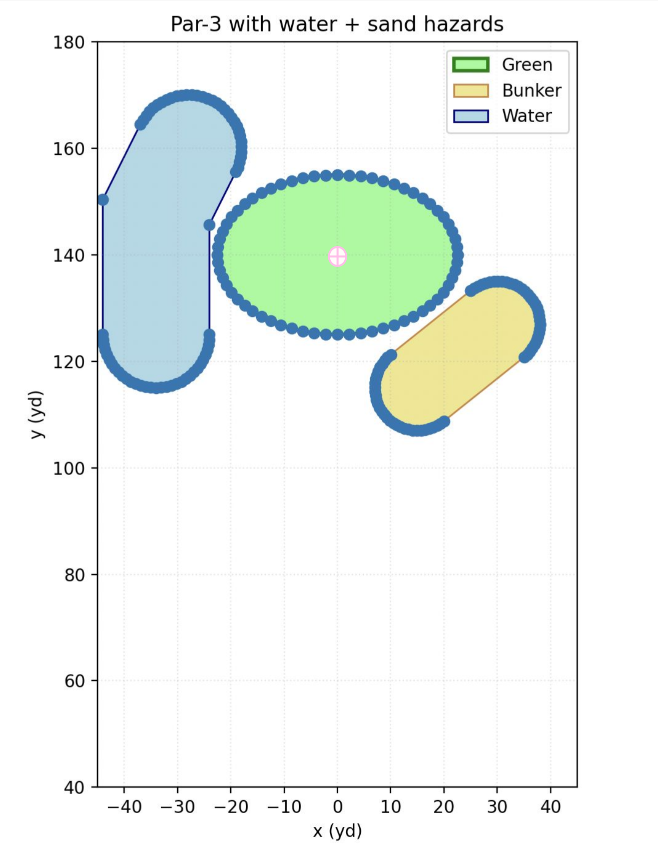


GPR Prediction Curves and Binned Averages by Lie Type
Predicted strokes to hole out
Distance to hole (yards)
3.50
3.25
3.00
2.75
2.50
2.25
2.00
1.75
0
10
20
30
40
50
Fairway
Rough
Sand


GPR Heatmaps by Lie Type
Fairway
Rough
Sand
X
X
2.0
2.2
2.4
2.6
2.8
3.0
3.2
Predicted strokes to hole out

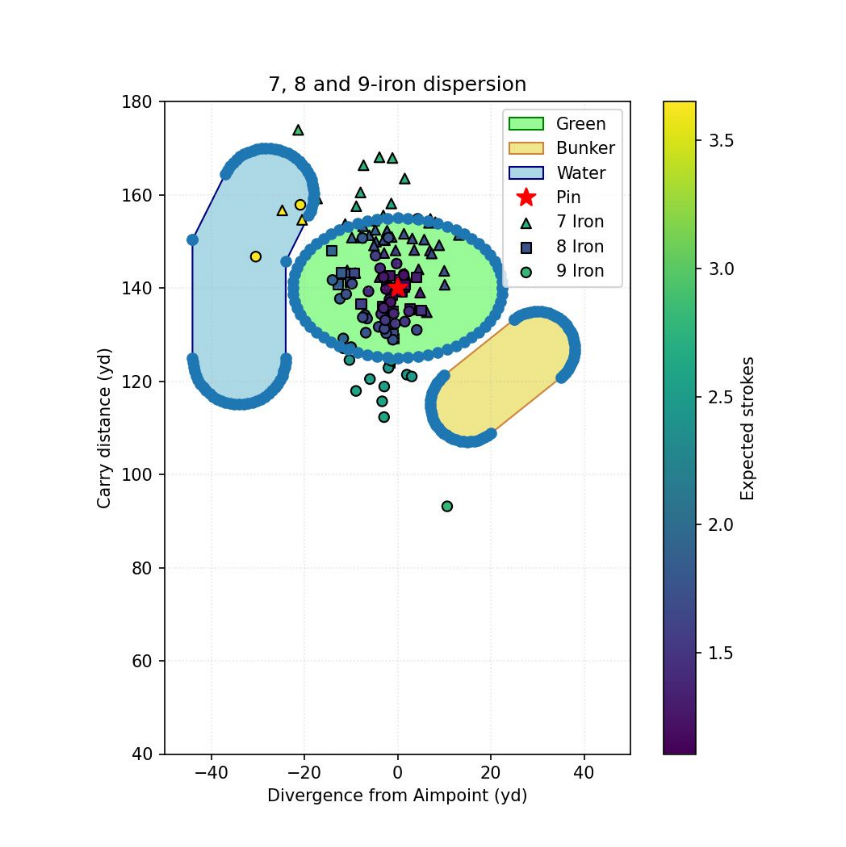
Par-3 Mock-up Hole

- Par-3 hole set at distance 140 yards
- Pin at (0, 140)
- Golfer starts at coordinate (0, 0)
- Hazards:
  - Water = +1 strokes
  - Sand
  - Rough (past 140 yards), behind green



Results from Example

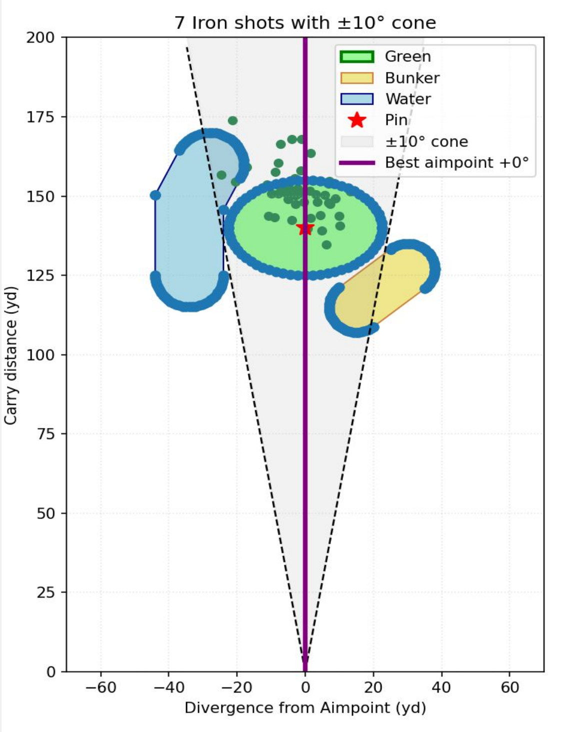
- Half-swing 7-iron metrics:
  - Mean: 2.04 ESHO
  - Median: 1.82 ESHO
- Full-swing 8-iron metrics:
  - Mean: 1.5 ESHO
  - Median: 1.38 ESHO
- Full-swing 9-iron metrics:
  - Mean: 1.9 ESHO
  - Median: 1.69 ESHO

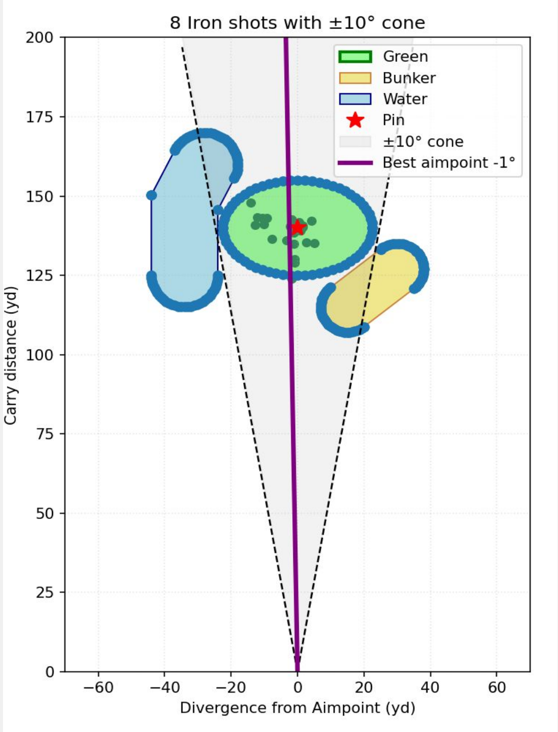


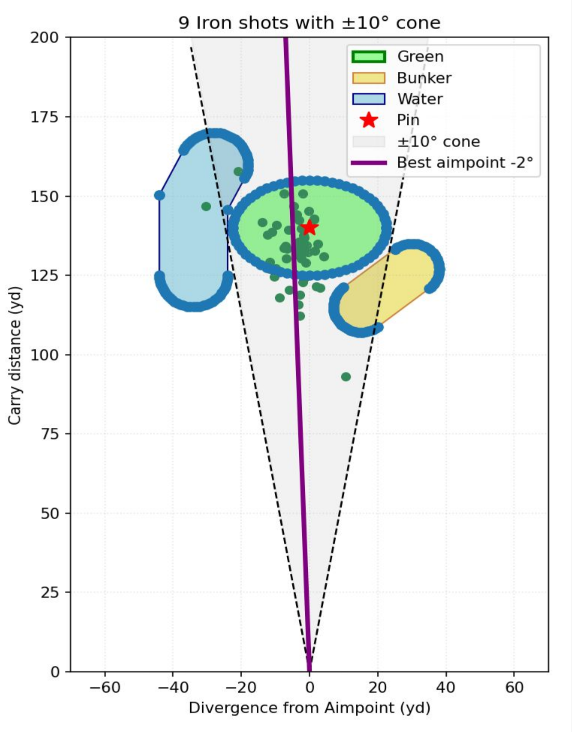
Varying Aimpoints

- Idea: Varying aimpoint degree away from pin location → players have tendencies! Don't aim straight at pin, instead, aim for error.
- -10° → +10°
  - Cone shaped
- Finding best aimpoint degree per each club
- Rotation matrix applied on each shot in accordance with Θ
  - Calculate ESHO on rotated (x,y) coordinates

Varying Aimpoints







Best: 0° (aim at pin)

Best: -1° (aim left of pin)

Best: -2° (aim left of pin)

Implementing Loss Function

- Idea: playoff scenarios where you need a certain score
  - Need a birdie, par, etc. to win

$$L = \frac{1}{n}\sum_{i=1}^{n} e^{(k*(s_i-t))}$$

- n = # of shots, s_i = ESHO for shot i, t = target score (birdie = 2), k = sharpness parameter (bigger k = harsher)

Implementing Loss Function

- Need Birdie (target = 2)
  - 7-iron: 155.495
  - 8-iron: 8.22
  - 9-iron: 141.185
- Need Par (target = 3)
  - 7-iron: 7.742
  - 8-iron: 0.409
  - 9-iron: 7.025
- Findings: need birdie or par → choose 8-iron for this hole

Thank you!